외계행성 대기
Exoplanetary Atmospheres

Lecture 6: Radiative Transfer IV — ine Formation, Scattering, Monte Carlo, and Approximate Solutio
2026 October 19 (Monday)

Based on: Chapter 6 of Seager, *Exoplanet Atmospheres* (2010)

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2026-10-17 10:15

Outline – Lecture 6

Line Formation and Radiative Transfer Solutions

An Introduction to Line Formation

Solutions I: Transmission and Thermal Emission

Solutions II: Scattering Atmospheres

Monte Carlo Radiative Transfer

Approximate Solutions: Rosseland, Eddington, and the Grey Atmosphere

Summary

Continuum, absorption lines, emission lines

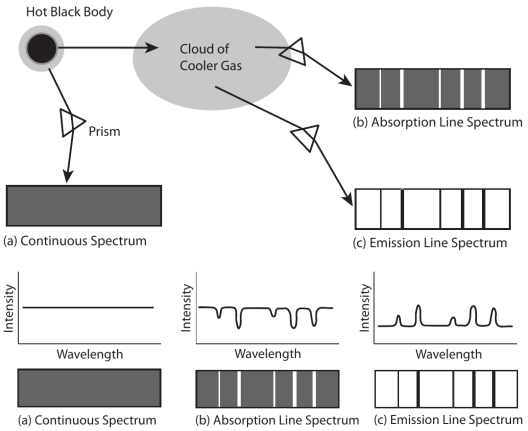


Figure 1: Origin of the three kinds of spectra (top) and what they look like over a narrow wavelength range (bottom). Seager (2010), Figs. 6.2 and 6.3.

- **Continuous spectrum:** a black body; the spectrum depends only on its temperature.
- **Absorption lines:** a cooler gas in front of the hot black body takes "bites" out of the continuum at the wavelengths its atoms and molecules absorb. The absorbed energy is re-emitted isotropically, so little of it returns to the line of sight.
- **Emission lines:** the same gas seen *without* the hot background. Only the re-emitted photons reach us, and they appear at exactly those wavelengths.
- What matters is not the gas alone but the gas *compared with what lies behind it*.

A layered atmosphere: the temperature gradient decides

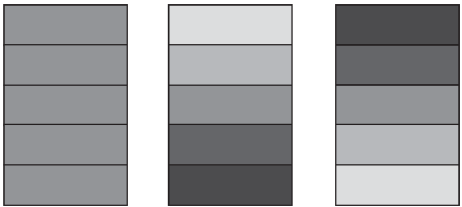


Figure 2: A five-layer atmosphere; darker means hotter. Left: isothermal. Middle: hot at depth, cooler above. Right: a temperature inversion. Seager (2010), Fig. 6.4.

Treat each layer as a homogeneous slab with its own temperature and the deepest layer as a black body.

- **Temperature decreasing outward:** every layer is a cool cloud in front of a hotter one $\Rightarrow$ an **absorption spectrum**. Each layer adds its own bites.
- **Temperature inversion** (hotter upward) $\Rightarrow$ an **emission spectrum**.
- **Isothermal:** no cooler gas in front, no hotter gas behind $\Rightarrow$ a black-body continuum with *no* features at all, however large the opacity.

The last point is counter-intuitive and worth proving (next section): spectral features measure temperature *contrasts*, not the mere presence of absorbers.

Transmission and scattered-light spectra

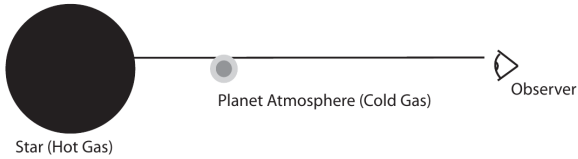


Figure 3: Origin of a transit transmission spectrum: hot stellar light through the cool planetary atmosphere. Seager (2010), Fig. 6.5.

- **Transmission:** the star is the hot black body, the planet's limb is the cool cloud ⇒ always an absorption spectrum. It is isolated by dividing the in-transit spectrum by the out-of-transit stellar spectrum.
- **Scattered light** (visible, near-IR): Seager prefers "scattered" to "reflected", because the light is refracted, diffracted and Rayleigh-scattered rather than bounced off a surface. To first order Earth's visible spectrum is sunlight that crossed the atmosphere *twice* (down to the ground or a cloud deck and back up), a weighted mix of clear and cloudy paths.

Isothermal Atmosphere: layer by layer

Take an n-layer LTE atmosphere whose deepest layer n is optically thick, so $I_n = B_n$. With Kirchhoff's law $\varepsilon_n = \kappa_n B_n$,

$$\mu \frac{\mathrm{d}I_n}{\mathrm{d}z} = \varepsilon_n - \kappa_n I_n = \kappa_n (B_n - B_n) = 0.$$

The intensity entering layer $n-1$ is therefore B_n, and in that layer

$$\mu \frac{\mathrm{d}I_{n-1}}{\mathrm{d}z} = \kappa_{n-1} (B_{n-1} - B_n) = 0 \qquad \text{because } T_{n-1} = T_n.$$

Repeating up to the top, $I_0 = \cdots = I_{n-1} = I_n = B$.

- Absorption and re-emission still happen in every layer, but they occur at the *same temperature* and hence at the same wavelengths: every photon removed is replaced by one of the same frequency.
- Note what the proof used: only the equality of temperatures, not the size of κ_ν. A strongly absorbing and a weakly absorbing wavelength give the same $I = B$.

Isothermal Atmosphere: from the formal solution

For an emitting atmosphere under **LTE conditions**, without scattering and with no radiation entering from above, the emergent intensity is (Lecture 3)

$$I(0,\mu,\nu) = \int_0^\infty B\big(T(\tau_\nu),\nu\big)\,\mathrm{e}^{-\tau_\nu/\mu}\,\frac{\mathrm{d}\tau_\nu}{\mu}.$$

If T is constant, B leaves the integral, and $\int_0^\infty \mathrm{e}^{-\tau/\mu}\,\mathrm{d}\tau/\mu = 1$, so

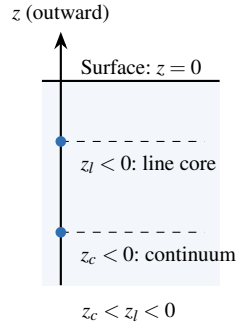
$$I(0,\mu,\nu) = B(T,\nu), \qquad F(0,\nu) = \pi B(T,\nu).$$

- The kernel $\mathrm{e}^{-\tau/\mu}/\mu$ is a *normalized weighting function*: the emergent intensity is a weighted average of B over depth, with weights concentrated near $\tau \sim \mu$.
- A weighted average of a constant is that constant. Only when B varies with depth can different wavelengths, weighted over different depths, come out different.

Non-isothermal Atmosphere: a temperature gradient creates lines

Compare a wavelength in the line core, ν_l, with one in the nearby continuum, ν_c. By definition $\kappa(\nu_l) > \kappa(\nu_c)$.

- From $\mathrm{d}\tau_\nu = -\kappa_\nu \,\mathrm{d}z$ (z increases outward), the same value $\tau_\nu \approx 1$ is reached *higher up* in the line core than in the continuum: $z_c < z_l < 0$ ($|z_l| < |z_c|$).
- The emergent intensity is essentially B at $\tau_\nu \approx \mu$ (Eddington–Barbier). So the line core samples higher layers, the continuum deeper ones.
- **Temperature decreasing outward:** $T(z_l) < T(z_c)$, so $I(\nu_l) < I(\nu_c)$ — an **absorption line**.
- **Temperature increasing outward (inversion):** $T(z_l) > T(z_c)$, so $I(\nu_l) > I(\nu_c)$ — an **emission line**.



A spectrum across a line is therefore a *scan in altitude*: the wings, with lower opacity, look deeper, and the core looks highest.[1]

[1] Seager's text at this point reads "$I(\tau_\nu, \mu, \nu_l) < I(\tau_\nu, \mu, \nu_l)$"; the right-hand side should be the continuum, ν_c.

Non-isothermal Atmosphere: the emergent flux as a depth-weighted B_ν

Integrating the emergent intensity over the outgoing hemisphere,

$$F(0,\nu) = 2\pi \int_0^1 \int_0^\infty B(\tau_\nu,\nu)\, \mathrm{e}^{-\tau_\nu/\mu}\, \mathrm{d}\tau_\nu\, \mathrm{d}\mu = 2\pi \int_0^\infty B(\tau_\nu,\nu)\, E_2(\tau_\nu)\, \mathrm{d}\tau_\nu,$$

using $\int_0^1 \mathrm{e}^{-\tau/\mu}\, \mathrm{d}\mu = E_2(\tau)$, the exponential integral $E_n(x) = \int_1^\infty t^{-n} \mathrm{e}^{-xt}\, \mathrm{d}t$.

- $2E_2(\tau)$ is the **flux weighting function**: it is normalized ($\int_0^\infty 2E_2\, \mathrm{d}\tau = 1$), equals 2 at the top and falls roughly as $\mathrm{e}^{-\tau}/\tau$ at depth.
- This is the working formula for thermal emission from an exoplanet without scattering: a black body in each layer plus the (much harder) opacities. Most retrieval codes evaluate exactly this sum.
- Written per unit $\log p$, the integrand $E_2(\tau)\, \mathrm{d}\tau/\mathrm{d}\log p$ is the **contribution function**: it shows which pressures a given wavelength actually probes.

Line formation in a toy atmosphere

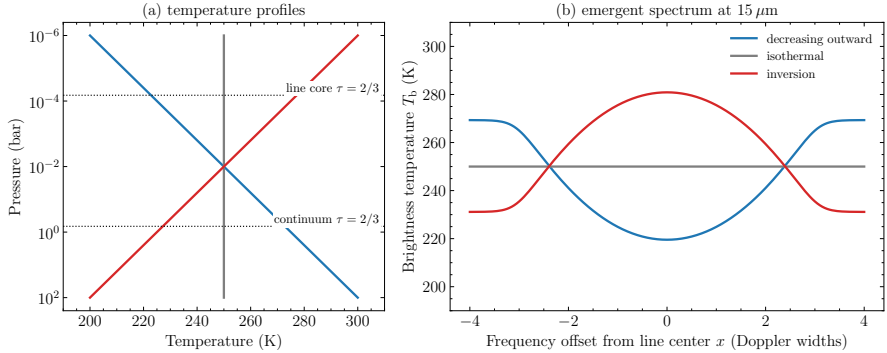


Figure 4: A line with a Gaussian opacity profile (peak 10^4 times the continuum) in three LTE atmospheres. Left: the temperature profiles and the pressures where $\tau = 2/3$ in the continuum and at line center. Right: the emergent flux as a brightness temperature. Computed with $F = 2\pi \int B E_2(\tau_\nu)\, \mathrm{d}\tau_\nu$; script `make_line_formation.py`.

The continuum reports T near 0.7 bar, the line core T near 10^{-4} bar; the isothermal case is featureless despite the huge opacity contrast.

Why a band has a floor: the minimum flux

A molecular absorption band is made of many closely spaced lines (Chapter 8 of Seager), so from here on we speak of an absorption *band*. Take an atmosphere whose temperature decreases outward.

- **The question:** what is the *minimum* flux in the band — and why is it not zero?
- There is always some radiation, however little, emerging at every wavelength.
- In LTE the size of the emergent flux depends on the temperature of the altitude where that flux *originates*.

So the question becomes: from which altitude does the flux at each wavelength come? The answer is set by the opacity, which varies strongly with wavelength — next slide. In an LTE atmosphere the flux at any wavelength is never zero; it is the Planck function of whatever layer that wavelength "sees".

Maximum and minimum flux: Earth's 15 μm CO_2 band

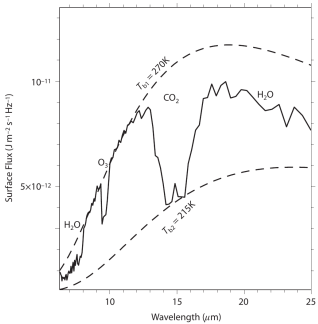


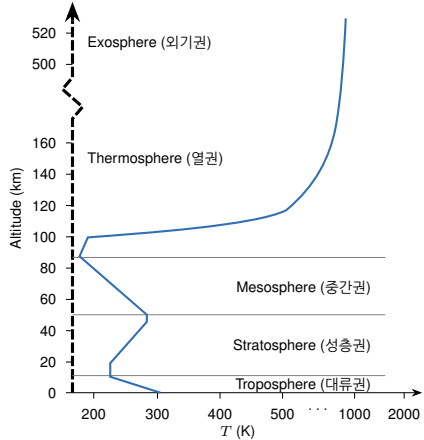
Figure 5: Earth's mid-infrared spectrum with black bodies at $T_{b1} = 270$ K and $T_{b2} = 215$ K. Reproduced from Seager (2010), Fig. 3.4.

- At a **low-opacity** wavelength the radiation can travel essentially unimpeded, so it emerges from *deep*, warm layers: this is the **maximum** flux, set by the surface or deep temperature.
- At a **high-opacity** wavelength the radiation is reprocessed in every layer on its way out; what finally escapes must have originated near the *top*, where it is cold: this is the **minimum** flux, set by the coldest layer the band can reach.
- Because opacity depends on wavelength, the spectrum *samples different temperatures* — in LTE, flux of different magnitude.
- **Earth:** between the bands the spectrum follows the warm $T_{b1} = 270$ K curve; the 15 μm CO_2 band bottoms out on the cold $T_{b2} = 215$ K curve, not at zero.
- The band depth therefore measures the *temperature range* between the continuum and band-center photospheres, not the amount of CO_2 as such. Brightness temperatures are a direct, if coarse, thermometer at different heights.

Reading Earth's temperature profile from its spectrum

- The deep O_3 (9.6 μm) and CO_2 (15 μm) absorption bands tell us that temperature decreases with height where they form: the **troposphere**, which holds most of the mass.
- At the very center of both bands there is a **narrow emission spike**. The core forms highest, so somewhere above the troposphere the temperature must rise again: the **stratospheric inversion**, heated by ozone absorbing solar UV.
- Why is the spike narrow? The opacity of a band scales with the number density of absorbers. Only at the band center is the opacity large enough for $\tau \approx 1$ to be reached in the low-density stratosphere; a little off-center, $\tau \approx 1$ drops back into the troposphere.
- The narrowness of the emission core tells us that the emission features are forming in regions of low density.

The width of an emission core thus encodes how high, and in how thin a layer, the inversion sits.



Earth's temperature profile. The exosphere extends to an altitude 500 km above Earth's surface. The ionosphere is a layer within the thermosphere. Redrawn after Seager (2010), Fig. 4.16.

Thermal inversions in hot Jupiters?

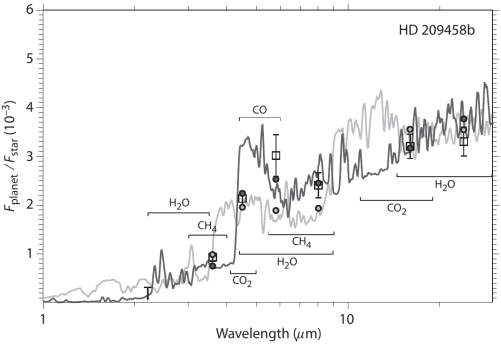


Figure 6: Historical interpretation of HD 209458b: Spitzer photometry (squares with error bars). Assuming H_2O and CO, the emission-only model (black) fits better than the absorption-only model (gray). Seager (2010), p. 112, Fig. 6.6; the inversion inference was revised by later observations.

- **Seager (2010):** secondary eclipses probe the permanent dayside of tidally locked hot Jupiters. Spitzer data suggested H_2O emission in some planets, absorption in others.
- **Earth:** weak emission cores within deep absorption bands indicate an inversion *above* the layers forming most of the absorption.
- **Initial HD 209458b interpretation:** strong emission without discernible absorption suggested an **on-average deep dayside inversion**, reaching the main feature-forming layers.
- **Update — HD 209458b:** HST/WFC3 resolved H_2O *absorption* (6.2σ). Together with Spitzer data, it supports **no inversion** over the probed $1–10^{-3}$ bar range (Line et al. 2016).
- **Update — WASP-121b:** resolved H_2O *emission* supports a stratospheric inversion (5σ; Evans et al. 2017).

Pure transmission: Lambert's law or Beer's law

In transmission there is neither emission nor scattering *into* the beam: $\varepsilon = 0$, hence $S = 0$, and

$$\mu \frac{\mathrm{d}I}{\mathrm{d}z} = -\kappa I \quad \Longrightarrow \quad I(z_f) = I(z_i) \exp\left[-\frac{1}{\mu}\int_{z_i}^{z_f} \kappa \, \mathrm{d}z'\right], \qquad I(\tau_\nu) = I(0)\, \mathrm{e}^{-\tau_\nu/\mu}.$$

In a transit the relevant quantity is the ratio of transmitted to incident stellar light along one ray, the stellar intensity cancelling:

$$\frac{F_{\mathrm{S},p,\mathrm{trans}}(\nu)}{F_{\mathrm{S},*,\mathrm{trans}}(\nu)} = \mathrm{e}^{-\tau_\nu},$$

where τ_ν is now the optical depth *along the chord* through the limb.

- This is one ray. The observed transit depth sums rays at all impact parameters through the annulus, each with its own chord optical depth (Lecture 3: $\tau_{\mathrm{chord}} \approx \sqrt{2\pi R_p/H}$ times the vertical one).
- Pressure broadening matters at depth, which is why the sum over altitudes cannot be skipped in a real calculation.

A transmission spectrum: HD 189733b

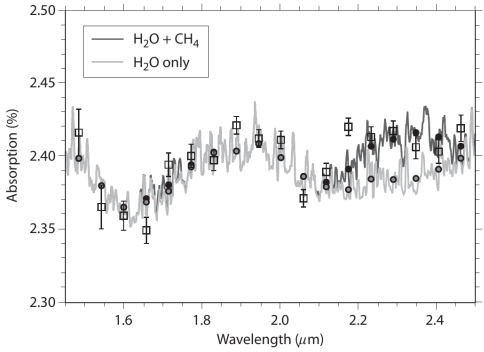


Figure 7: HST/NICMOS transmission spectrophotometry of HD 189733b (open squares with error bars). Curves: H_2O only (gray) or $H_2O + CH_4$ (black). Open circles and filled circles are, respectively, the H_2O-only and $H_2O + CH_4$ model predictions averaged over the observational wavelength bins, accounting for the instrument response. Seager (2010), Fig. 6.7.

- The signal is $\sim 5 \times 10^{-4}$ of the stellar flux, as estimated in Chapter 3: the area of an annulus about $5H$ tall relative to the stellar disk.
- Atoms and molecules detected in transmission in hot Jupiters include Na, K, H_2O, CO, CO_2 and (with JWST) SO_2.
- This particular CH_4 detection was later disputed: the NICMOS systematics had been underestimated (Gibson et al. 2011). A useful reminder that a feature is only as good as the instrument model behind it.

Seager's exercise 3(c) asks when thermal emission from the planet would matter in transmission: when the planet's own emission along the chord is not negligible compared with the transmitted starlight, i.e. for very hot planets observed in the infrared.

The emergent flux

The emergent surface flux follows from the flux definition (Chapter 2) applied to the outgoing hemisphere, $F = 2\pi \int_0^1 \mu\, I(0,\mu)\, \mathrm{d}\mu$ (no incoming radiation at the top in the thermal infrared):

$$\boxed{F(0,\nu) = 2\pi \int_0^1 \int_0^\infty S(\tau_\nu, \mu, \nu)\, \mathrm{e}^{-\tau_\nu/\mu}\, \mathrm{d}\tau_\nu\, \mathrm{d}\mu}$$

- If the source function $S(\tau, \mu, \nu)$ is *known*, the emergent intensity and the emergent flux — the planet's spectrum — can be computed directly from these two integrals. The μ integral of $\mathrm{e}^{-\tau/\mu}$ produces the exponential integral $E_2(\tau)$, so $F(0,\nu) = 2\pi \int_0^\infty S(\tau_\nu) E_2(\tau_\nu)\, \mathrm{d}\tau_\nu$ for an isotropic S.
- For $S = B(T(\tau))$ (pure absorption, LTE) this is the standard *forward model* of an emission spectrum: choose $T(\tau_\nu)$, i.e. a temperature–pressure profile and an opacity function, and integrate.
- In many cases, however, this straightforward solution is not possible. The main complication is that the source function itself depends on $I(\tau, \mu, \nu)$, the quantity we are trying to solve for.

Thermal emission: the formal solution with $S = B$

At infrared wavelengths molecular absorption dominates and scattering can be neglected: $\kappa = \alpha_{\rm abs}$ (absorption only), and in LTE $S_{\rm therm} = \varepsilon/\kappa = B(\tau_\nu, \nu)$. The equation $\mu\, \mathrm{d}I/\mathrm{d}\tau_\nu = I - B$ has the solution

$$I(0, \mu, \nu) = \int_0^\infty B(\tau_\nu, \nu)\, \mathrm{e}^{-\tau_\nu/\mu}\, \frac{\mathrm{d}\tau_\nu}{\mu}, \qquad \boxed{F(0, \nu) = 2\pi \int_0^1 \int_0^\infty B\, \mathrm{e}^{-\tau_\nu/\mu}\, \mathrm{d}\tau_\nu\, \mathrm{d}\mu}.$$

- "Powerfully simple": each layer is a black body; the only hard ingredient is the opacity, which fixes $\tau_\nu(z)$.
- Exact in LTE without scattering, given $T(z)$. This makes it the workhorse for secondary-eclipse and directly imaged spectra.
- Thermal emission peaks near the Planck maximum: ~ 0.8–$1\,\mu$m for 2000–3000 K ultra-hot Jupiters, $\sim 15\,\mu$m for Earth, longer for colder planets.
- Next slide shows data for one example, the Spitzer and Hubble Space Telescope observations of HD 189733b.

An emission spectrum: HD 189733b

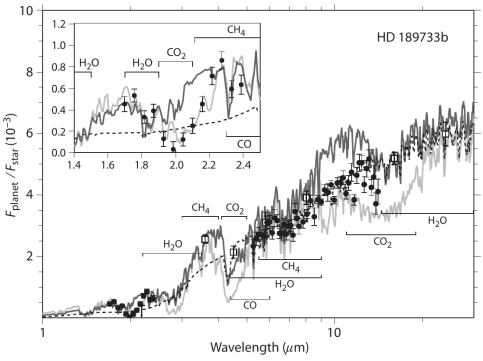


Figure 8: Secondary-eclipse data for HD 189733b: Spitzer photometry (squares), Spitzer/IRS (circles), HST/NICMOS (inset), with best-fit models. Seager (2010), Fig. 6.8.

- The planet/star contrast reaches $\sim 5 \times 10^{-3}$ in the mid-infrared, ten times the transmission signal.
- Order of magnitude (Chapter 3, Rayleigh–Jeans limit):

$$\frac{F_p}{F_*} \approx \frac{T_{\mathrm{b},p}}{T_{\mathrm{b},*}} \left(\frac{R_p}{R_*}\right)^2 .$$

  With $R_p/R_* = 0.155$ and $T_{\mathrm{b},*} \approx 5000\,\mathrm{K}$, a contrast of 3.4×10^{-3} at $8\,\mu\mathrm{m}$ implies $T_{\mathrm{b},p} \approx 700\,\mathrm{K}$; the full Planck ratio gives $\approx 1300\,\mathrm{K}$, close to the measured dayside value.
- Absorption features of H_2O and CO/CO_2: temperature decreasing with height on this dayside.

A constant source function

We can solve the RT equation under the assumption of a constant source function with altitude. Take S independent of depth between an upper level ($\tau_{\min}$) and a lower one ($\tau_{\max}$); along a vertical ray,

$$I(\tau_{\min}) = I(\tau_{\max})\,\mathrm{e}^{-(\tau_{\max}-\tau_{\min})} + S\left[1-\mathrm{e}^{-(\tau_{\max}-\tau_{\min})}\right].$$

The first term is what survives of the light entering from below, the second what the slab emits along the path.

- **Optically thick slab** (deep in the atmosphere), $\tau_{\max}-\tau_{\min} \gg 1$: $I = S$. The incoming light is forgotten; only local conditions matter, and in LTE $I = S = B$: a featureless continuum. *Lines do not form at large optical depth.*

$$I(0,\nu) = S(\nu).$$

- **Optically thin top layer**, $\tau_{\min} = 0$, $\tau_{\max} < 1$, so $\mathrm{e}^{-\tau} \approx 1-\tau$:

$$I(0,\nu) = I(\tau_{\max},\nu) - \tau_{\max}(\nu)\left[I(\tau_{\max},\nu) - S(\nu)\right].$$

  [Absorption Line Spectrum] The layer *removes* light if the background is brighter than its source function (i.e., $I(\tau_{\max},\nu) > S(\nu)$).

  [Emission Line Spectrum] The layer *adds* light if it is fainter (i.e., $I(\tau_{\max},\nu) < S(\nu)$), in proportion to its optical depth $\tau_{\max}(\nu)$.

Absorption or emission: the four cases

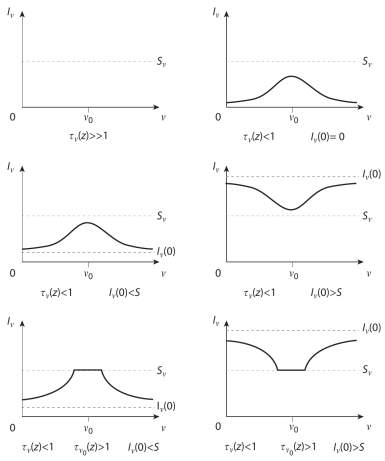


Figure 9: Line formation for a source function constant with depth and frequency. Seager (2010), Fig. 6.9, after Rutten.

Read the panels with $I(\tau_{\max}) \equiv I_\nu^0$, the light from behind, and $\tau_\nu(z)$ the layer's optical depth, largest at line center ν_0:

- **Top left** — thick at all ν: $I = S$, no line.
- **Top right** — thin, no backlight ($I_\nu^0 = 0$): emission line, strength $\propto \tau_\nu$.
- **Middle left** — thin, $I_\nu^0 < S$: emission line on top of the background.
- **Middle right** — thin, $I_\nu^0 > S$: absorption line.
- **Bottom** — thick at line center only ($\tau_{\nu_0} > 1$): the core *saturates* at $I = S$, giving a flat-topped emission or flat-bottomed absorption line.

Absorption lines require light from behind that is *brighter* than the layer's own source function — the same statement as "cool gas in front of a hot body".

A linear source function: Eddington–Barbier approximation

Let $S(\tau_\nu) = a_\nu + b_\nu\,\tau_\nu$ (independent of μ). The formal solution gives exactly

$$I_\nu(0,\mu) = a_\nu + b_\nu\mu, \qquad F_\nu(0) = 2\pi\int_0^1 (a_\nu + b_\nu\mu)\,\mu\,\mathrm{d}\mu = \pi\left(a_\nu + \tfrac{2}{3}b_\nu\right). \qquad \boxed{F_\nu(0) = \pi S(\tau_\nu = 2/3).}$$

- The emergent **flux** is characteristic of $\tau_\nu \approx 2/3$; the **intensity** along the vertical ray, $I(0,1) = S(\tau_\nu = 1)$. This is the Eddington–Barbier relation.[2]
- It does not require the absence of scattering, only a source function that is (close to) linear in τ.
- Because τ_ν depends on frequency, so does the physical height of $\tau_\nu = 2/3$: this is the mathematical form of **"the line core forms higher (outer layer)"**.
- The same $I = a_\nu + b_\nu\mu$ gives $\boxed{K(\tau_\nu,\nu) = J(\tau_\nu,\nu)/3}$ for the second and zeroth moments — the **Eddington approximation**, which reappears as the third Eddington coefficient of the two-stream method.

[2] Seager writes $F(0,\nu) = S(\tau_\nu = 2/3)$ and $F(0,\nu) = S(\tau_\nu = 1)$ (Eqs. 6.32–6.33); the first lacks the factor π, and the second is the vertical intensity rather than a flux.

A Scattering Atmosphere: Why scattering makes everything harder

Absorption and thermal emission tie the radiation to the *local* temperature. Scattering does not:

- A scattered photon can travel far before it is absorbed. The long-distance photon travel by scattering therefore **couples distant parts of the atmosphere**, and even the open boundary at the top influences the field deep inside.
- The source function now contains the radiation field itself,

$$S_{\text{scat}}(\tau,\mu,\phi) = \frac{\omega_0}{4\pi}\int_{4\pi} P(\mu,\phi;\mu',\phi')\,I(\tau,\mu',\phi')\,\mathrm{d}\Omega',$$

  so the RT equation becomes an **integro-differential equation**: I appears on both sides, once differentiated and once integrated over all directions.
- Few analytic solutions exist; classical texts spend many pages even on isotropic scattering (Chandrasekhar 1960; Sobolev 1975).

Here ω_0 ($\tilde{\omega}$ in Seager) is the single-scattering albedo and P the single-scattering phase function, normalized as $\int_{4\pi} P\,\mathrm{d}\Omega/4\pi = 1$ (Lecture 3).

Direct beam and diffuse field

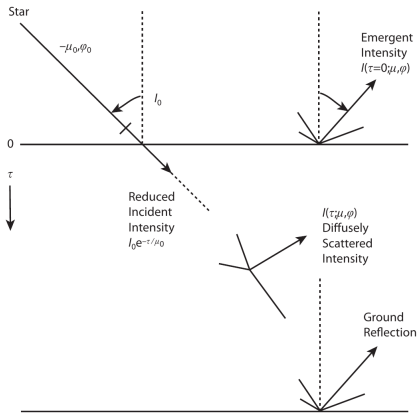


Figure 10: The attenuated direct beam from the star and the diffusely scattered intensity. Seager (2010), Fig. 6.10, after Chamberlain & Hunten.

Starlight arrives as a parallel beam from direction (μ_0, ϕ_0):

$$I_*(0,\mu,\phi) = I_0\,\delta(\mu-\mu_0)\,\delta(\phi-\phi_0).$$

The incident flux follows from angular integration (Seager, p. 120):

$$\begin{aligned} F_*(0,\mu,\nu) &= \int_0^{2\pi}\int_{-1}^{0} \mu I_0\,\delta(\mu-\mu_0)\,\delta(\phi-\phi_0)\,\mathrm{d}\mu\,\mathrm{d}\phi \\ &= \mu_0 I_0. \end{aligned}$$

In this signed-flux equation, incoming $\mu_0 < 0$; the incident-flux magnitude is $|\mu_0| I_0$. For attenuation below, μ_0 denotes the positive magnitude.

Unscattered, the beam is attenuated as

$$I_{\mathrm{dir}}(\tau) = I_0\,\mathrm{e}^{-\tau/\mu_0}\,\delta(\mu-\mu_0)\,\delta(\phi-\phi_0).$$

It is convenient to split the field into this **direct beam**, known analytically, and the **diffuse field** I of photons scattered at least once, which is what we must solve for.

Scattering RT: an angularly coupled problem

Let τ_ν increase inward and $\mu > 0$ denote outgoing rays. For coherent scattering, the source receives photons from all directions:

$$\boxed{\mu \frac{\mathrm{d}I_\nu(\tau_\nu,\Omega)}{\mathrm{d}\tau_\nu} = I_\nu(\tau_\nu,\Omega) - S_\nu(\tau_\nu,\Omega)}$$

$$S_{\mathrm{scat},\nu}(\tau_\nu,\Omega) = \frac{\omega_0}{4\pi}\int_{4\pi} P_\nu(\Omega;\Omega')I_\nu(\tau_\nu,\Omega')\,\mathrm{d}\Omega'.$$

- $\Omega = (\mu,\phi)$, $\mathrm{d}\Omega' = \mathrm{d}\mu'\,\mathrm{d}\phi'$; P is normalized by $\int P\,\mathrm{d}\Omega/4\pi = 1$. Pure scattering has $\omega_0 = 1$.
- Scattering removes photons from one ray and adds photons from other rays. Unlike an LTE thermal source $B_\nu(T)$, the scattering source depends on the *unknown radiation field*.
- The equation is therefore **integro-differential**: every direction is coupled to the others.

Notation correction: use $I - S$, as in Seager Eq. 6.1; the source is an integral over solid angle, without a flux factor μ'. The printed Eqs. 6.38–6.40 have inconsistent signs/angular measures.

Separate the direct beam from diffuse scattering

Write $I_{\rm tot} = I_{\rm dir} + I_{\rm diff}$. Here $\mu_0 > 0$ is the *magnitude* of the incoming cosine, so the stellar beam travels in direction $\Omega_0 = (-\mu_0, \phi_0)$.

$$S_{{\rm single},\nu}(\tau_\nu, \Omega) = \frac{\omega_0 I_0}{4\pi}\, \mathrm{e}^{-\tau_\nu/\mu_0} P_\nu(\Omega; \Omega_0),$$
$$S_{{\rm diff},\nu}(\tau_\nu, \Omega) = \frac{\omega_0}{4\pi} \int_{4\pi} P_\nu(\Omega; \Omega') I_{{\rm diff},\nu}(\tau_\nu, \Omega')\, \mathrm{d}\Omega',$$

- The direct beam is known analytically. Its source term describes the **first scattering of stellar photons**; the diffuse term accounts for photons that scatter again.
- For an outgoing ray, a layer of total depth τ_b gives

$$\boxed{I_\nu(0, \mu) = I_\nu(\tau_b, \mu)\mathrm{e}^{-\tau_b/\mu} + \frac{1}{\mu}\int_0^{\tau_b} S_\nu(t, \mu)\mathrm{e}^{-t/\mu}\, \mathrm{d}t}$$

- This formal solution is still valid, but $S_{\rm diff}$ contains the unknown $I_{\rm diff}$: it must be solved self-consistently.

Seager Eqs. 6.41–6.42 rewritten with the required $1/\mu$ and an explicit lower-boundary term. For stellar scattering alone, $S = S_{\rm single} + S_{\rm diff}$; thermal emission adds $(1 - \omega_0) B_\nu$.

Single scattering in a thin layer

Optically thin cirrus or aerosol layers high in the atmosphere can be dominated by one scattering of the incident stellar beam. Set $S = S_{\rm single}$ and neglect illumination from below:

$$I_\nu(0,\mu) = \frac{\omega_0 I_0 P_\nu}{4\pi\mu}\int_0^{\tau_b} \mathrm{e}^{-t/\mu_0}\mathrm{e}^{-t/\mu}\,\mathrm{d}t = \frac{\omega_0\mu_0 I_0 P_\nu}{4\pi(\mu+\mu_0)}\left[1-\mathrm{e}^{-\tau_b(1/\mu+1/\mu_0)}\right].$$

The exponentials account for attenuation **before** and **after** scattering. For $\tau_b/\mu \ll 1$ and $\tau_b/\mu_0 \ll 1$,

$$\boxed{R(\mu,\phi;\mu_0,\phi_0) \equiv \frac{\pi I_\nu(0,\mu,\phi)}{\mu_0 I_0} \simeq \frac{\omega_0\tau_b}{4\mu\mu_0} P_\nu(\mu,\phi;-\mu_0,\phi_0)}$$

- R is Seager's **dimensionless bidirectional reflectance** (BRDF): a Lambert-normalized reflected intensity. On Earth, the BRDF can be determined from satellite measurements.
- At fixed geometry, albedo and phase function, $R \propto \tau_b$. No isotropic-scattering or two-stream assumption is needed.
- A small *vertical* depth is insufficient near grazing rays: the relevant incoming and outgoing *slant* depths must be small.

Seager Eq. 6.45 correction: the incoming attenuation is e^{-t/μ_0}, not $\mathrm{e}^{-t}\mu_0$. The finite-depth expression is exact for the single-scattering term, not for unrestricted multiple scattering.

Multiple scattering and Earth's spectrum

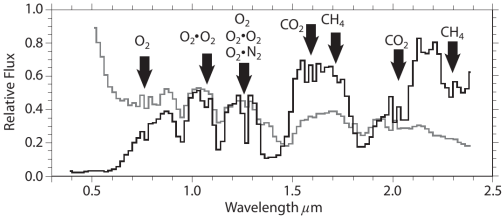


Figure 11: Earth as an exoplanet. Both spectra were observed by telescopes on Earth, using the Moon as a reflector. Gray (earthshine): Sun → Earth (scattering/reflection) → Moon → telescope on Earth. Black (lunar eclipse): Sun → Earth's atmospheric limb (transmission/refraction) → Moon → telescope on Earth. The long slant paths in transmission and the cloud/surface-weighted paths in scattered light produce different band strengths. Labels identify O_2, O_2–O_2 and O_2–N_2 collision-induced absorption, CO_2 and CH_4. Relative fluxes emphasize spectral shapes, rather than absolute planet/star contrasts. Seager (2010), Fig. 6.11; Pallé et al. (2009).

- Once repeated scattering matters, the source and radiation field are coupled; a simple single-scattering solution is no longer sufficient.
- **Adding–doubling:** calculate reflection and transmission for a thin homogeneous layer. Combine layers, including repeated reflections between them; doubling builds a thick slab efficiently.
- Different homogeneous layers can be added to represent a **vertically inhomogeneous** atmosphere.
- Two hot Jupiters had visible geometric-albedo upper limits below 0.15, compared with Jupiter's ~ 0.5; this helped explain the difficulty of scattered-light detection.
- Earth's scattered and transmitted spectra probe different paths and weight clouds and gases differently. Their molecular-band strengths therefore differ (left figure).

Radiative transfer in cloudy atmospheres

The same RT equations apply to clouds. The numerical difficulty is the **sharp opacity change at an optically thick cloud boundary**. Seager (p. 125) summarizes several practical treatments:

1. **Cloud top as a boundary:** place the effective atmospheric surface at the top of an opaque cloud and prescribe its emerging radiation, instead of solving through the cloud interior.
2. **Partial cloud coverage:** combine cloudy and clear columns, possibly with different cloud-top heights:

$$\boxed{I_{\text{disk}} \simeq f_{\text{cl}} I_{\text{cloudy}} + (1 - f_{\text{cl}}) I_{\text{clear}}}.$$

   Here f_{cl} is the projected-area cloud fraction; the columns are assumed radiatively independent.
3. **Smooth the boundary:** replace an abrupt cloud edge with an exponential density decrease for numerical convenience.
4. **Approximate cloud scattering analytically:** prescribe the radiation emerging from a cloud layer without solving its full internal transfer.

For realistic, nonuniform **3D cloud structures**, **Monte Carlo RT** approach is the most accurate way to deal with scattering from clouds.

Monte Carlo RT: follow photon packets

Traditional RT solves for the intensity field; Monte Carlo RT samples **individual photon-packet histories**. Their ensemble estimates the same radiation field.

- A packet represents many photons at frequency ν, with a position, direction and energy; polarization can also be tracked.
- Draw the next interaction distance, absorption or scattering, and the new direction from the appropriate probability laws. Follow each packet until it is **absorbed or escapes**.
- **Strength:** irregular geometry and spatially varying properties, including clumpy 3D clouds, can be treated directly.
- **Cost:** many histories are needed to reduce sampling noise; optically thick regions require many scattering events. Seager discusses simulations with millions of packets or more.

Scope here: time-independent, monochromatic, coherent scattering. The method can be applied to 3D media, thermal emission and temperature-structure calculations.

Sampling a free path: inverse cumulative probability

Let $p(x)$ be a normalized probability density on $[a,b]$. Draw ζ uniformly in $(0,1)$ and invert its cumulative distribution:

$$\boxed{\zeta = \int_a^{x_0} p(x)\,\mathrm{d}x, \qquad \int_a^b p(x)\,\mathrm{d}x = 1}.$$

The probability of no interaction after a path optical depth t is e^{-t}. Thus the next-interaction probability density is

$$p(t)\,\mathrm{d}t = \mathrm{e}^{-t}\,\mathrm{d}t, \qquad t \geq 0.$$

$$\zeta = 1 - \mathrm{e}^{-t} \quad \Longrightarrow \quad \boxed{t = -\ln(1-\zeta)}.$$

- t is optical depth *along the packet's ray*, not vertical depth. Convert it into a distance s using the local extinction: $t = \int_0^s \chi_v(\mathbf{r} + s'\hat{\mathbf{n}})\,\mathrm{d}s'$.
- In a homogeneous medium, $s = t/\chi_v$. If a boundary is closer than the sampled interaction, the packet reaches that boundary first.

The photon-packet algorithm

1. **Initialize.** Launch a packet from the stellar source; set its entry position, direction, frequency and energy.
2. **Propagate.** Sample $t = -\ln(1-\zeta_1)$ and find the corresponding path length. If the packet exits first, record it; otherwise move to the interaction point.
3. **Choose the interaction.** With a new random number, scatter if $\zeta_2 < \omega_0$ and absorb otherwise:

$$\boxed{\Pr(\text{scatter}) = \omega_0, \qquad \Pr(\text{absorb}) = 1-\omega_0}.$$

4. **Scatter and repeat.** Draw a new outgoing direction from $p(\Omega \mid \Omega') = P(\Omega;\Omega')/(4\pi)$; update polarization if included, then return to step 2. Coherent scattering preserves ν.
5. **Absorb or escape.** Absorption deposits the packet energy and ends its contribution to this monochromatic scattering calculation. Escaping packets contribute to the emergent radiation.

Absorption does not destroy energy: re-emission at thermal wavelengths is outside this scattering-only example. A surface encounter requires an appropriate absorption/reflection boundary condition.

From escaping packets to observables and errors

Bin escaped packets by direction, exit location and frequency. Their energy, normalized to the injected luminosity and bin geometry, gives the emergent flux; directional bins describe phase and inclination dependence (and polarization, if tracked).

- **Equal-energy packets:** if each packet carries energy $E_{\rm pkt}$ over a normalization time Δt, then $L_{\rm bin} = N_{\rm bin} E_{\rm pkt}/\Delta t$. Conversion to intensity or observer flux must include the relevant area, solid angle and distance.
- **Sampling error:** Seager uses Poisson counting errors. For independent, equal-weight packets and well-populated bins,

$$\boxed{\sigma_N \simeq \sqrt{N_{\rm bin}}, \qquad \frac{\sigma_F}{F} \simeq \frac{1}{\sqrt{N_{\rm bin}}}}.$$

  Halving the statistical error requires roughly four times as many packets in that bin.
- **Numerical check:** in a steady scattering-only run, escaped energy plus absorbed energy should match injected energy.
- Monte Carlo naturally handles complex clouds, but noise and long random walks make optically thick calculations expensive. Weighted-packet methods require their own variance estimates.

Approximate solutions

How to solve the radiative transfer equation:

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = -\alpha_\nu^{\mathrm{ext}}\,(I_\nu - S_\nu), \qquad S_\nu = (1-\varepsilon_\nu)J_\nu + \varepsilon_\nu B_\nu, \quad \text{with} \quad \varepsilon_\nu = \frac{\alpha_\nu^{\mathrm{abs}}}{\alpha_\nu^{\mathrm{ext}}}.$$

(Here $\varepsilon_\nu = 1-\omega_0$: the fraction of extinction that is true absorption, so $S = (1-\omega_0)B + \omega_0 J$ as before.)

We will learn two approximations to solve the equation in a **plane-parallel** medium, primarily developed for stellar atmospheres:

- the **Rosseland** approximation (radiative diffusion),
- the **Eddington** approximation.

Together they give the temperature structure of a *grey* atmosphere and the limb darkening of the Sun, and they lead to the Milne solution met in the next lecture.

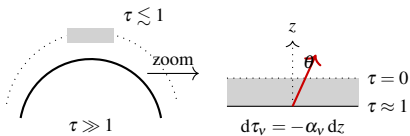
Plane-parallel atmosphere

Stellar atmosphere.

- The atmosphere of a star is its outer region, which determines the properties of the radiative flux that emanates from its surface.
- Computing its structure needs the detailed monochromatic opacity and powerful numerical codes; but with appropriate approximations interesting results follow analytically.

Plane-parallel atmosphere.

- We ignore the curvature of the underlying body (frequency independent opacity). Suitable when the atmosphere is a thin layer, "locally virtually flat".
- The optical depth is a function of the vertical distance z rather than the distance s along the ray; it increases downward, with $\tau = 0$ at the top: $\mathrm{d}\tau_\nu = -\alpha_\nu \, \mathrm{d}z$.



(1) Radiative diffusion: the Rosseland approximation

Imagine a plane-parallel medium in which ρ and T depend only on the depth z. Along a ray, $\mathrm{d}s = \mathrm{d}z/\cos\theta = \mathrm{d}z/\mu$, so

$$\mu \frac{\partial I_\nu(z,\mu)}{\partial z} = -\alpha_\nu^{\mathrm{ext}}(I_\nu - S_\nu) \quad \Longrightarrow \quad I_\nu(z,\mu) = S_\nu - \frac{\mu}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial I_\nu}{\partial z}.$$

- **"Zeroth" approximation.** Deep in the material, *all quantities change slowly on the scale of a mean free path* $\ell = 1/\alpha^{\mathrm{ext}}$, so the derivative term is very small:

$$I_\nu^{(0)}(z,\mu) \approx S_\nu^{(0)}(T) \quad \therefore \quad J_\nu^{(0)} = S_\nu^{(0)} \text{ and } I_\nu^{(0)} = S_\nu^{(0)} = B_\nu \quad (\mathrm{LTE}),$$

  independent of the angle.

- **"First" approximation.** Insert $I^{(0)}$ in the derivative term:

$$I_\nu^{(1)}(z,\mu) \approx S_\nu^{(0)} - \frac{\mu}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial I_\nu^{(0)}}{\partial z} = B_\nu(T) - \frac{\mu}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu(T)}{\partial z} \quad \to \text{ linear in } \mu.$$

  With $\mathrm{d}\tau_\nu = -\alpha_\nu^{\mathrm{ext}}\mathrm{d}z$ this is $I_\nu \simeq B_\nu + \mu\,\mathrm{d}B_\nu/\mathrm{d}\tau_\nu$.

What the first approximation means

$$\boxed{I_\nu(\tau_\nu,\mu) \simeq B_\nu(\tau_\nu) + \mu\,\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}} \qquad (\tau_\nu \text{ increasing inward})$$

- The leading term is **isotropic and Planckian**, $I_\nu \simeq S_\nu \simeq B_\nu$; as $\tau \to \infty$, $I = S = B$. A small anisotropy $\propto \mu$ carries the net flux.
- Its moments give $K = J/3$ *exactly* — the Eddington relation of the next part, and Heng's **third Eddington coefficient** $\varepsilon_3 = K/J = 1/3$. In the two-stream picture of Lecture 7 the up- and down-going streams merge here: net and total fluxes are no longer distinct, and scattering is effectively isotropic ($g_0 = 0$).
- This is a **deep-interior** approximation, not a solution for the escaping radiation at $\tau_\nu \lesssim 1$.
- Large *scattering* depth alone does not guarantee thermalization: $S_\nu \to B_\nu$ requires enough true absorption ($\tau_* = \sqrt{3\varepsilon}\,\tau \gg 1$, below).

Seager's Eq. (6.49) prints $I \approx B - \mu\,\mathrm{d}B/\mathrm{d}\tau$; with τ increasing inward the sign must be $+$ to agree with the transfer equation and with her Eq. (6.51).

The net flux in the first approximation

The angle-independent part of the intensity does not contribute to the net flux along z:

$$F_\nu(z) = \int I_\nu^{(1)}(z,\mu)\cos\theta\,\mathrm{d}\Omega = 2\pi\int_{-1}^{+1} I_\nu^{(1)}(z,\mu)\,\mu\,\mathrm{d}\mu$$
$$= -\frac{2\pi}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu}{\partial z}\int_{-1}^{+1}\mu^2\,\mathrm{d}\mu = -\frac{4\pi}{3\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu(T)}{\partial T}\frac{\partial T}{\partial z} = \frac{4\pi}{3}\frac{\mathrm{d}B_\nu}{\mathrm{d}\tau_\nu}.$$

Integrating over frequency, the **total flux** is

$$F(z) = \int_0^\infty F_\nu(z)\,\mathrm{d}\nu = -\frac{4\pi}{3}\frac{\partial T}{\partial z}\int_0^\infty \frac{1}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu.$$

Define the **Rosseland mean absorption coefficient**

$$\boxed{\frac{1}{\alpha_R} \equiv \frac{\displaystyle\int_0^\infty \frac{1}{\alpha_\nu^{\mathrm{ext}}}\frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu}{\displaystyle\int_0^\infty \frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu}}$$

The equation of radiative diffusion

Use $\int_0^\infty \frac{\partial B_\nu}{\partial T}\,\mathrm{d}\nu = \frac{\partial}{\partial T}\int_0^\infty B_\nu\,\mathrm{d}\nu = \frac{\partial(\sigma T^4/\pi)}{\partial T} = \frac{4\sigma T^3}{\pi}$. Then the Rosseland approximation to the radiative flux is

$$\boxed{F(z) = -\frac{16\sigma T^3}{3\alpha_R}\frac{\partial T}{\partial z} \;\rightarrow\; -\chi\nabla T}$$

the **equation of radiative diffusion**, one of the fundamental equations for stellar interiors.

- It has the form of Fourier's law of heat conduction, with the **radiative conductivity** $\chi = 16\sigma T^3/3\alpha_R$: flux travels *down* the temperature gradient.
- **Per unit mass** (Heng): $\alpha_R = \rho\kappa_\mathrm{R}$. With hydrostatic equilibrium, $\mathrm{d}P = -\rho g\,\mathrm{d}z$,

$$F = \frac{16\,\sigma\,g\,T^3}{3\,\kappa_\mathrm{R}}\frac{\mathrm{d}T}{\mathrm{d}P}.$$

Notation: α is extinction per unit length — Seager's κ, Heng's α_e; $\kappa_\mathrm{R} = \alpha_R/\rho$ is the Rosseland mean per unit mass.

Why a harmonic mean

Three ways to average 1 and 100 (the Pythagorean means):

$$\text{arithmetic } \frac{1+100}{2} = 50.5, \qquad \text{geometric } (1\times 100)^{1/2} = 10, \qquad \text{harmonic } \frac{2}{1+1/100} \approx 2.$$

- The harmonic mean is dominated by the *smallest* value. The Rosseland mean averages $1/\alpha_\nu^{\mathrm{ext}}$, so the frequencies at which the extinction is **small** (transparent) dominate.
- In an optically thick medium radiation leaks out through these **windows**, just as current in parallel resistors flows through the smallest resistance.
- The weighting function $\partial B_\nu/\partial T$ resembles the Planck function but peaks at $h\nu_{\max} = 3.8\,k_{\mathrm{B}}T$ instead of $2.8\,k_{\mathrm{B}}T$.
- Hence a small amount of an opacity source that fills a window (a continuum absorber, a cloud) can change α_R and the deep thermal structure far more than a strong line in an already opaque band.

(2) The Eddington approximation

In the Eddington approximation the intensities are assumed to approach *isotropy*, not necessarily their thermal values (in the Rosseland approximation they approach the Planck function at large depth). Near isotropy can be introduced by taking the intensity linear in μ (frequency suppressed):

$$I(\tau,\mu) = a(\tau) + b(\tau)\,\mu.$$

The first three moments are

$$\text{mean intensity:}\quad J \equiv \tfrac{1}{2}\int_{-1}^{+1} I\,\mathrm{d}\mu = a,$$

$$\text{flux:}\quad H \equiv \tfrac{1}{2}\int_{-1}^{+1} \mu I\,\mathrm{d}\mu = \frac{b}{3},$$

$$\text{radiation pressure:}\quad K \equiv \tfrac{1}{2}\int_{-1}^{+1} \mu^2 I\,\mathrm{d}\mu = \frac{a}{3},$$

so that

$$\boxed{K = \tfrac{1}{3}J} \qquad \text{the } \textbf{Eddington approximation}.$$

Compare the isotropic radiation field, $p = \frac{1}{3}u$, with $p \equiv \frac{1}{c}\int I\cos^2\theta\,\mathrm{d}\Omega$ and $u = \frac{4}{c}I$.

The Eddington approximation: the radiative diffusion equation

The optical depth and the transfer equation are $\mathrm{d}\tau(z) \equiv -\alpha^{\mathrm{ext}}\mathrm{d}z$ and $\mu\,\partial I/\partial\tau = I - S$. The source function does not depend on μ, because $S = (1-\varepsilon)J + \varepsilon B$. Integrating over μ:

$$\tfrac{1}{2}\int_{-1}^{+1}\mathrm{d}\mu\left(\mu\frac{\partial I}{\partial\tau} = I - S\right) \;\rightarrow\; \frac{\partial H}{\partial\tau} = J - S, \qquad \tfrac{1}{2}\int_{-1}^{+1}\mathrm{d}\mu\,\mu\left(\mu\frac{\partial I}{\partial\tau} = I - S\right) \;\rightarrow\; \frac{\partial K}{\partial\tau} = H \;\rightarrow\; \frac{1}{3}\frac{\partial J}{\partial\tau} = H.$$

The two equations combine into

$$\boxed{\frac{1}{3}\frac{\partial^2 J}{\partial\tau^2} = J - S = \varepsilon\,(J - B)}$$

Define a new optical depth

$$\tau_* \equiv \sqrt{3\varepsilon}\,\tau = \sqrt{3\tau_{\mathrm{abs}}(\tau_{\mathrm{abs}} + \tau_{\mathrm{sca}})},$$

and the radiative equation becomes

$$\boxed{\frac{\partial^2 J}{\partial\tau_*^2} = J - B}$$

the **radiative diffusion equation**: given the temperature structure $B(\tau)$ of the medium, it can be solved for J.

The grey atmosphere

If **the opacity is independent of frequency**, illustrative and insightful results can be found; a stellar atmosphere with such an opacity spectrum is a *grey atmosphere*.

- The radiation field still has a frequency distribution, since matter emits a spectrum determined by its temperature.
- We look for the temperature profile (versus depth) and the flux that emanates from a grey atmosphere.
- It is now more convenient to let τ increase *inward* (independent of frequency):

$$\boxed{\mu \frac{\mathrm{d}I_\nu}{\mathrm{d}\tau} = I_\nu - S_\nu}$$

- **(1)** Integrating over all frequencies,

$$\boxed{\mu \frac{\mathrm{d}I}{\mathrm{d}\tau} = I - S} \qquad I = \int_0^\infty I_\nu \,\mathrm{d}\nu, \quad S = \int_0^\infty S_\nu \,\mathrm{d}\nu.$$

Radiative equilibrium

If there are no energy sources or sinks in the atmosphere, the energy absorbed by a layer equals the energy it emits. In the plane-parallel approximation this means the energy generated in the interior passes out as a *constant energy flux* through the outer layers.

$$\iint \alpha_\nu I_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega = \iint j_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega \qquad \text{(energy absorbed or scattered } = \text{ energy radiated)}$$

In the **grey atmosphere**, with the definition of the source function ($j_\nu = \alpha_\nu S_\nu$),

$$\iint \alpha_\nu I_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega = \iint \alpha_\nu S_\nu \,\mathrm{d}\nu\,\mathrm{d}\Omega \quad \rightarrow \quad \alpha \int J_\nu \,\mathrm{d}\nu = \alpha \int S_\nu \,\mathrm{d}\nu \quad \Longrightarrow \quad \boxed{J = S}$$

Radiative equilibrium: the moment equations

(2) Average the first equation over the solid angle:

$$\tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu \left(\mu \frac{\mathrm{d}I_\nu}{\mathrm{d}\tau}\right) = \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, I_\nu - \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, S_\nu \;\Rightarrow\; \frac{\mathrm{d}H}{\mathrm{d}\tau} = J - S \xrightarrow{J=S} \boxed{\frac{\mathrm{d}H}{\mathrm{d}\tau} = 0, \quad H = \text{constant}}$$

(3) Multiply the first equation by μ and average over the solid angle:

$$\tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu \left(\mu^2 \frac{\mathrm{d}I}{\mathrm{d}\tau}\right) = \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, (\mu I) - \tfrac{1}{2}\int_{-1}^{1} \mathrm{d}\mu\, (\mu S) \;\Rightarrow\; \boxed{\frac{\mathrm{d}K}{\mathrm{d}\tau} = H} \quad \text{(isotropic source function).}$$

Integrating gives the radiation pressure as a function of optical depth,

$$\boxed{K(\tau) = H(\tau + c)} \qquad (Hc \text{ is the constant of integration}).$$

Using the **Eddington approximation**, $K = J/3$, the mean intensity is

$$\boxed{J(\tau) = 3H(\tau + c)} \qquad \text{with } c = \tfrac{2}{3}, \text{ shown on the next slide.}$$

Derivation of the constant of integration

Solve $\mu\,\mathrm{d}I/\mathrm{d}\tau = I - S$ formally. For $\mu > 0$ (outgoing) multiply by $e^{-\tau/\mu}$ and integrate from $\tau' = \infty$ to τ; for $\mu < 0$ (ingoing) integrate from $\tau' = 0$, where nothing comes in:

$$\mu > 0:\ I(\tau,\mu) = \int_{\tau}^{\infty} e^{-(\tau'-\tau)/\mu} S \frac{\mathrm{d}\tau'}{\mu}, \qquad \mu < 0:\ I(\tau,\mu) = \int_{0}^{\tau} e^{-(\tau-\tau')/\mu} S \frac{\mathrm{d}\tau'}{|\mu|}.$$

At the surface, for $\mu > 0$ and with $S = J = 3H(\tau + c)$,

$$I(0,\mu) = \int_0^{\infty} e^{-\tau'/\mu} S \frac{\mathrm{d}\tau'}{\mu} = 3H \int_0^{\infty} e^{-\tau'/\mu} (\tau' + c) \frac{\mathrm{d}\tau'}{\mu} \quad \Longrightarrow \quad \boxed{I(0,\mu) = 3H(\mu + c)}$$

and $I(0,\mu) = 0$ for $\mu < 0$. Now use the definition of the outward flux H at $\tau = 0$:

$$H = \tfrac{1}{2}\int_{-1}^{1} \mu I(0,\mu)\,\mathrm{d}\mu = \tfrac{1}{2}\int_{0}^{1} \mu\, I(0,\mu)\,\mathrm{d}\mu = \frac{3H}{2}\left(\frac{1}{3} + \frac{c}{2}\right) \quad \Longrightarrow \quad \boxed{c = \frac{2}{3}}$$

Limb darkening

- Observations show that the intensity at the center of the solar disk is larger than near its limb: **limb darkening**.
- The local temperature increases with depth, and the radiation leaving the star comes from $\tau \approx 1$ along the line of sight. **The layers responsible for the radiation are hotter at the center than near the limb.**
- The Eddington limb-darkening law can be estimated from the grey-atmosphere results (next slide).

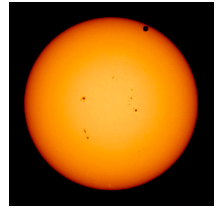
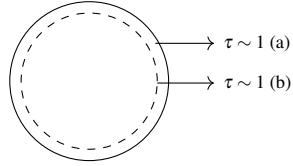


Image: Brocken Inaglory.



$T_a > T_b \Rightarrow S_a > S_b$

Limb darkening: the Eddington law

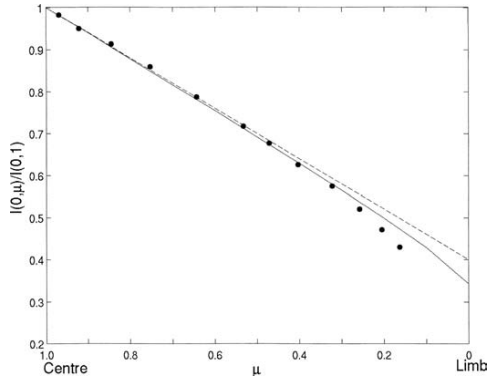
The intensity at the surface of the star is, with $S = J = 3H\,(\tau + \frac{2}{3})$,

$$I(\tau = 0, \mu) = \int_0^\infty S(\tau)\, e^{-\tau/\mu}\, \frac{\mathrm{d}\tau}{\mu}$$
$$= \int_0^\infty 3H\Big(\tau + \frac{2}{3}\Big) e^{-\tau/\mu}\, \mathrm{d}(\tau/\mu) = 3H\Big(\mu + \frac{2}{3}\Big).$$

The direction toward the center of the disk is $\theta = 0$, $\mu = 1$; the limb corresponds to $\mu < 1$. The ratio of the intensities is

$$\boxed{\frac{I(0,\mu)}{I(0,1)} = \frac{3}{5}\Big(\mu + \frac{2}{3}\Big)}$$

the **Eddington limb-darkening law**. At the limb ($\mu = 0$) the ratio is 0.4: the intensity of the grey-atmosphere approximation at the limb is 40% of its value at the center of the disk.



Dots: observation. Dashed: Eddington approximation. Solid: exact grey-atmosphere solution; $\lambda = 5485$ Å (Pierce et al. 1950).

The temperature profile of a grey atmosphere

From the definition of the effective temperature,

$$H = \frac{1}{4\pi}\int \cos\theta\, I\, \mathrm{d}\Omega = \frac{1}{4\pi}F = \frac{1}{4\pi}\sigma_{\mathrm{SB}} T_{\mathrm{eff}}^4.$$

The relation between the intensity and the outward flux, $F = \pi J$, gives

$$J(\tau) = \frac{1}{\pi}F(\tau) = \frac{1}{\pi}\sigma_{\mathrm{SB}} T^4(\tau),$$

where T is the effective temperature at τ. Combining with $J(\tau) = 3H\left(\tau + \frac{2}{3}\right)$,

$$T^4 = \frac{3}{4}T_{\mathrm{eff}}^4\left(\tau + \frac{2}{3}\right) \quad \Longrightarrow \quad \boxed{T = T_{\mathrm{eff}}\left[\frac{3}{4}\left(\tau + \frac{2}{3}\right)\right]^{1/4}}$$

This tells us how the temperature varies inside a grey atmosphere.

- At $\tau = 2/3$, $T = T_{\mathrm{eff}}$: the temperature at the photosphere is the effective temperature.
- The same form follows from radiative diffusion alone: **Milne's solution** (next slide).

Milne's solution

Deep in the atmosphere the flux is carried outward by radiation and, in radiative equilibrium, is constant with depth; set it to the internal heat flux, $F = \sigma_{\mathrm{SB}} T_{\mathrm{int}}^4$. With the Rosseland optical depth $\mathrm{d}\tau_R = -\alpha_R \,\mathrm{d}z$, the diffusion equation reads

$$\sigma_{\mathrm{SB}} T_{\mathrm{int}}^4 = \frac{16\sigma_{\mathrm{SB}} T^3}{3} \frac{\mathrm{d}T}{\mathrm{d}\tau_R} = \frac{4\sigma_{\mathrm{SB}}}{3} \frac{\mathrm{d}T^4}{\mathrm{d}\tau_R},$$

and integrating over τ_R,

$$\boxed{T^4 = \frac{3}{4} T_{\mathrm{int}}^4 \left(\tau_R + C\right)}$$

with $C \sim 1$ a constant of integration.

- This is **Milne's solution**, first derived for stars. Diffusion alone cannot fix C, because it does not hold near the surface. The Eddington approximation with the boundary condition of no incoming radiation gives $C = 2/3$: exactly the grey-atmosphere profile of the previous slide, with $T_{\mathrm{int}} = T_{\mathrm{eff}}$.
- At $\tau_R \approx 1$ the observer measures $T \approx T_{\mathrm{int}}$: for the Sun, $T_{\mathrm{int}} \approx 5800\,\mathrm{K}$ is the photospheric temperature. Temperature rises as $\tau_R^{1/4}$ with depth.
- In an irradiated planet this internal part is joined by the absorbed starlight; combining the two gives the temperature–pressure profiles of a later lecture.

Boundary conditions: the two-stream approximation

To solve the second-order equation for J we need two boundary conditions. One way is the **two-stream approximation**: the whole radiation field is represented by radiation at just two angles, $\mu = \pm\mu_0$,

$$\boxed{I(\tau,\mu) = I^+(\tau)\,\delta(\mu-\mu_0) + I^-(\tau)\,\delta(\mu+\mu_0)}$$

The two terms are the outward and inward intensities. The three moments are

$$J = \tfrac{1}{2}\big(I^+ + I^-\big), \qquad H = \tfrac{1}{2}\mu_0\big(I^+ - I^-\big), \qquad K = \tfrac{1}{2}\mu_0^2\big(I^+ + I^-\big),$$

and $K = \frac{1}{3}J$ requires $\mu_0 = 1/\sqrt{3}$ ($\theta_0 = \cos^{-1}\mu_0 = 54.74°$). Using $H = \frac{1}{3}\,\partial J/\partial\tau$ we obtain

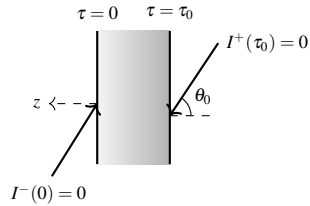
$$\boxed{I^\pm = J \pm \frac{1}{\sqrt{3}}\frac{\partial J}{\partial \tau}}$$

Boundary conditions: a slab with no incident radiation

Let the medium extend from $\tau = 0$ to $\tau = \tau_0$ with no incident radiation, $I^-(0) = 0$ and $I^+(\tau_0) = 0$; with $I^{\pm} = J \pm \frac{1}{\sqrt{3}} \partial J / \partial \tau$ this gives

$$\boxed{\begin{aligned} \frac{1}{\sqrt{3}} \frac{\partial J}{\partial \tau} &= J \quad \text{at } \tau = 0, \\ \frac{1}{\sqrt{3}} \frac{\partial J}{\partial \tau} &= -J \quad \text{at } \tau = \tau_0. \end{aligned}}$$

Note the generalization to N streams, in which the field is represented at N angles.



Take-home messages

Line formation. Spectral features measure temperature contrasts: an optically thick isothermal LTE atmosphere emits a featureless black body. Opaque wavelengths form high, transparent ones deep; temperature falling outward gives absorption, an inversion gives emission. Emergent flux $\approx \pi S(\tau_\nu = 2/3)$.

Solutions. Transmission obeys Beer's law along the chord; thermal emission is $F = 2\pi \int B E_2 \, \mathrm{d}\tau$ without scattering.

Scattering and Monte Carlo. Scattering couples directions; thin layers give $R \propto \omega_0 \tau P/(4\mu\mu_0)$. Clouds require boundary treatments or explicit scattering calculations. Sample photon paths and interactions; escaped packets give observables with counting noise $\propto N^{-1/2}$.

Approximate solutions. Rosseland (deep layers): $F = -(16\sigma T^3/3\alpha_R)\,\mathrm{d}T/\mathrm{d}z$ with the harmonic mean α_R, which with constant flux gives Milne's $T^4 = \frac{3}{4} T_{\mathrm{int}}^4 (\tau_R + C)$. Eddington: $K = J/3$, which in a grey atmosphere gives $J = 3H(\tau + \frac{2}{3})$, $T^4 = \frac{3}{4} T_{\mathrm{eff}}^4 (\tau + \frac{2}{3})$ and the limb-darkening law $I(\mu)/I(1) = \frac{3}{5}(\mu + \frac{2}{3})$.

Next. Practical solutions of the transfer equation, clouds and retrieval, the two-stream approximation and diffusion (Lecture 7).

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